

STA314 Assignment 1

Due Date: October 7th, 2026, 11:59pm

Instructions:

- LATEX or R Markdown is recommended but not required for submission. However, it is expected that responses are neatly organized, especially when handling equations and R codes.
- For problems requiring R programming, include necessary codes and outputs only. If your codes are complicated, make sure you comment on essential implementations.
- Submit through Crowdmark

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1. You will derive the Bias-Variance tradeoff formula from the lecture. Assume that we have a regression model

$$Y = f(X) + \epsilon,$$

where ϵ is independent of X and $\mathbb{E}(\epsilon) = 0, \mathbb{E}(\epsilon^2) = \sigma^2$. Assume that the training data $(x_1, y_1), \dots, (x_n, y_n)$ are used to construct an estimate of f , denoted by $\hat{f}$. Given a new random vector (X, Y) (independent of the training data):

- (a) Show that

$$\mathbb{E} \left[(f(x) - \hat{f}(x))^2 | X = x \right] = \text{Var} \left(\hat{f}(x) \right) + \left[\mathbb{E}[\hat{f}(x)] - f(x) \right]^2$$

- (b) Show that

$$\mathbb{E} \left[(Y - \hat{f}(x))^2 | X = x \right] = \text{Var} \left(\hat{f}(x) \right) + \left[\mathbb{E}[\hat{f}(x)] - f(x) \right]^2 + \sigma^2$$

2. It is well-known that ridge regression tends to give similar coefficient values to correlated variables, whereas the lasso may give quite different coefficient values to correlated variables. We will now explore this property in a very simple setting.

Suppose that $n = 2, p = 2, x_{11} = x_{12}, x_{21} = x_{22}$. Furthermore, suppose that $y_1 + y_2 = 0$ and $x_{11} + x_{21} = 0$ and $x_{12} + x_{22} = 0$, so that the estimate for the intercept in a least squares, ridge regression, or lasso model is zero: $\hat{\beta}_0 = 0$.

- (a) Write out the ridge regression optimization problem in this setting.
 - (b) Argue that in this setting, the ridge coefficient estimates satisfy $\hat{\beta}_1 = \hat{\beta}_2$
 - (c) Write out the lasso optimization problem in this setting.
 - (d) Argue that in this setting, the lasso coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are not unique—in other words, there are many possible solutions to the optimization problem in (c). Describe these solutions.
3. For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer in 1 – 2 sentences.
 - (a) The sample size n is extremely large, and the number of predictors p is small.
 - (b) The number of predictors p is extremely large, and the number of observations n is small.
 - (c) The variance of the error terms, i.e. $\text{Var}(\epsilon)$, is extremely high.

- (d) The relationship between the predictors and response is highly non-linear.
4. In this question, you will explore real-world applications of statistical learning methods. You may use resources such as *Google Scholar* to identify relevant research papers. For each application, **provide complete citations for the papers you select**.
- (a) Find two real-world applications in which **classification** was used to address a research or practical problem. For each application:
 - briefly describe the problem being studied;
 - state whether the primary goal is **inference** or **prediction**;
 - identify the response variable and the predictors used.
 - (b) Find two real-world applications in which **regression** was used to address a research or practical problem. For each application:
 - briefly describe the problem being studied;
 - state whether the primary goal is **inference** or **prediction**;
 - identify the response variable and the predictors used.
 - (c) Find two real-world applications in which **cluster analysis** was used to address a research or practical problem. For each application:
 - briefly describe the problem being studied;
 - describe what observations or objects were being clustered;
 - explain the purpose of performing the cluster analysis in the context of the application.
5. In this question, we will explore the per capita crime rate, `crim`, in the Boston dataset from the ISLR2 package in R. To load the dataset, use the code below:

```
library(ISLR2)
data(Boston)
data <- Boston
```

Treat ‘`crim`’ as the response variable and the remaining variables as potential predictors. Click [this link](#) to find more information on the dataset, which will aid you in your understanding of the problem.

- (a) Fit models using each of the following regression methods discussed in lecture:
 - best subset selection;
 - forward subset selection;
 - the Lasso;
 - ridge regression.

Summarize and discuss your results; do not include large amounts of unprocessed R output. Clearly describe any choices you make when fitting or selecting models.

- (b) Based on your analysis in part (a), propose a model (or set of models) that appears to perform well on this dataset, and justify your choice. Model performance should be evaluated using validation set error, cross-validation, or another reasonable approach, rather than training error. Clearly explain and justify the approach you use to evaluate model performance.
- (c) Do your chosen model(s) involve all of the predictors in the dataset? Explain why or why not.